

Fig. 26.50 Final alignment of tailstock spindle to bed ways in horizontal plane.

RECOMMENDED STANDARDS

Tool Room lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
0 to .0005"	0 to .0005"	0 to .0005"

(Forward at end of spindle when fully extended)

It makes no difference where the tailstock is placed on the ways for these operations, since the bed should now be in perfect condition. Ordinarily, one part of the bed ways is used as the position from which to conduct the alignment tests. Another portion is coated with compound and utilized as a spotting template. This practice increases speed because much clean up time is eliminated.

Two consecutive tests are conducted in attaining OBJECTIVE NO. 1. Both utilize the line of travel of the carriage along the bed ways as the medium whereby the axis of the tailstock spindle is tested for parallelism with the outer inverted V-ways of the bed.

To expedite the work, the apron is attached to the carriage. It takes but a few minutes, and then the handwheel may be used to move the member. This procedure is preferable to pushing the carriage by hand, because both the speed of traverse and the stopping point are more accurately controlled. (It is unnecessary to mount the lead screw to attach the apron.)

The first test consists of checking whether the axis of the fully extended tailstock spindle is parallel to the outside inverted V-ways of the bed in the horizontal plane. To conduct the survey a DIAL INDICATOR is mounted on the carriage, or more specifically, on the compound slide rest because this member is provided with a T-slot to which a suitable fixture may be conveniently bolted. The DIAL INDICATOR is positioned to bear against the near side of the extended spindle at the horizon-

tal diameter. (The tailstock spindle is clamped in the fully extended position.) To obtain a reading, the carriage is moved along the bed ways. The alignment of the tailstock spindle in the horizontal plane is considered acceptable if the unilateral tolerance indicated in Fig. 26.51 is not exceeded.

The second test consists of checking whether the axis of the fully extended tailstock spindle is parallel to the outer inverted V-ways of the bed, in the vertical plane. The spindle remains in the same fully extended position occupied for the previous test, but now the DIAL INDICATOR is adjusted so as to ride the top of the spindle at the vertical diameter. By moving the carriage on the inverted V-ways of the bed another reading can be taken. The alignment of the tailstock spindle in the vertical plane is acceptable if the unilateral tolerance shown in Fig. 26.51 is not exceeded.

In the recommended standards of both tests, working pressure is a factor. For example, in testing horizontally, it is the pressure of the tool against the work during a cut which must be taken into account. When testing vertically, it is the weight of the work piece for which allowance is made. If the unilateral tolerance is exceeded in either test, then the V-slide (1) and the flat slide (2) of the tailstock shown in Fig. 26.44 must be scraped so that the new alignment will conform to the tolerances given.

After determining the amount of error and its direction, the tailstock is removed to the portion of the bed ways reserved for spotting purposes. In doing this, the member must not be slid. Instead, it should be lifted and then placed carefully on the coated portion of the ways. Transfer of the film is made in the usual manner.

Scraping is most efficiently performed at a work bench. Before beginning this operation, a little reflection should enable the operator to determine how to scrape the tailstock V-slide so as to "throw" the axis of the spindle in the proper direction, thereby correcting the error disclosed in the two tests. The amount of metal to be removed from the tailstock slides at this time should be slight, since all the OBJECTIVES, previously prescribed for these surfaces, have been designed so that only a light touching up would be necessary during the Final Alignment.

During the initial stages of testing, the tailstock may simply be placed in position on the bed. But in the final stages, it should be clamped to the bed so that the tests will be conducted under circumstances similar to actual operating conditions.

This concludes the scraping operations on the tailstock member, although two additional tests must be conducted with the spindle before the member can be accepted for use.

Sec. 26.56 The Tailstock Spindle Taper Hole Alignment.

Briefly, the purpose of the following tests is to make certain that the tapered spindle hole has been properly machined, reamed, or ground so that the axis of the tapered hole is "in-line" with the axis of the lathe, or more specifically, "in-line" with the axis of the tailstock spindle.

To prove the accuracy of the alignment of the tapered hole in the tailstock spindle, two tests are conducted consecutively. Strictly speaking, the following tests do not concern scraping operations, but they are, nonetheless, vital to the successful alignment of this lathe member. When the accuracy of the tapered hole falls below the recommended standards, the tailstock spindle hole must be reworked, or a new spindle inserted.

Preliminary to conducting these tests it is necessary to make up a test bar. The bar is made of steel, hardened and accurately ground to fit the tapered hole in the spindle. It should have a straight portion of suitable length, usually 12". On some lathes, the test bar prepared for the tailstock can be used later on for the headstock, provided the tapered hole is identical in both members. When it is dissimilar, a separate bar is required for each member.

CAUTION:

It is not good practice to "bush" a test bar in order to adapt it to a larger hole.

Having prepared the bar, insert it into the tapered hole in the spindle. Make certain that the contacting surfaces are perfectly clean. Foreign matter may prevent the two tapers from seating accurately and

this leads invariably to unreliable readings. It is also obvious that the two tapers must be identical. For additional information on this point, refer to Sec. 15.4.

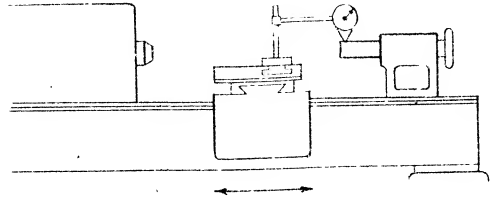


Fig. 26.51 Final alignment of tailstock spindle to bed ways in vertical plane.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
0 to .0005"	0 to .0008"	0 to .0015"

(High at end of spindle when fully extended)

The first test, diagrammed in Fig. 26.52 determines if the axis of the tapered hole in the spindle is "in-line" with the axis of the tailstock spindle in the horizontal plane. A DIAL INDICATOR is mounted on the carriage and adjusted against the near side of the test bar at the horizontal diameter. A reading is taken, as the carriage is moved forward. If the plus or minus tolerance shown in the Figure is not exceeded, the axis of the tapered hole in the spindle, is correctly aligned to the axis of the spindle in the horizontal plane.

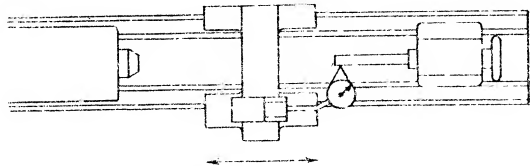


Fig. 26.52 Alignment of taper hole of tailstock in horizontal plane.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
0 to (+) .0005"	0 to (+) .0008"	0 to (+) .0015"

(At end of 12 inch test bar)

The next test which is diagrammed in Fig. 26.53 discloses if the axis of the

tapered hole in the spindle is "in-line" with the axis of the spindle in the vertical plane. The only change required for the set up in this test from the previous one, is to locate the DIAL button at the top of the test bar at the vertical diameter. A reading is taken as the carriage is traversed. If the unilateral tolerance indicated in the recommended standards is not exceeded, then the axis of the tapered hole is "in-line" with the axis of the spindle in the vertical plane.

In case the two tests show that the tapered hole is inaccurate and exceeds the limits allowed, the spindle must be removed from the tailstock. It must then be replaced with a new and accurate spindle. In lieu of this, the tapered hole of the defective spindle can be reworked. However, since this matter pertains to machine shop practice it will not be further discussed.

To reassure the critical reader, it will be emphasized that, finding a tailstock spindle with a defective tapered hole will not affect the alignment of the previously scraped bearing surfaces of the tailstock. Replacement of the spindle or re boring the taper hole will, however, be called for.

Sec. 26.57

The Head Stock of the Lathe

The fixed member of the lathe which carries the work spindle supporting the live center, and the necessary mechanism to drive the work, is called the head stock. It is a solid one piece casting, as shown in Fig. 26.54. The member has two bearing surfaces which are scraped, namely:

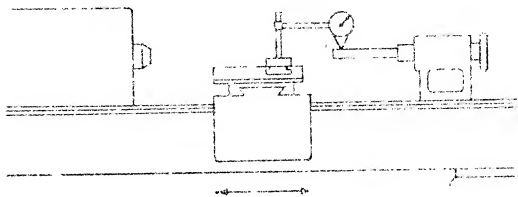


Fig. 26.53 Alignment of taper hole of tailstock in vertical plane.

RECOMMENDED STANDARDS

Tool Room	12" to 18" inc.	20" to 30" inc.
Lathes	Engine Lathes	Engine Lathes
0 to .0005"	0 to .0005"	0 to .0015"

(High at end of 12 inch test bar)

1. The V-slide
2. The flat slide

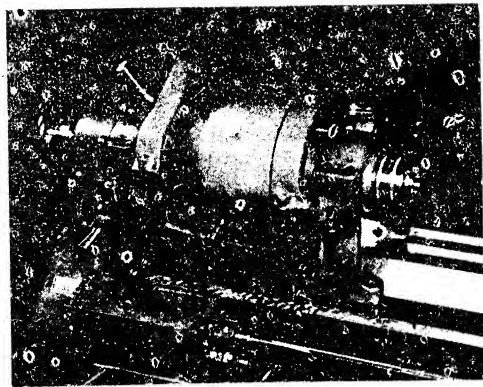


PLATE 31. Close-up of lathe headstock mounted on ways of bed. (Courtesy - South Bend Lathe Works.)

All spotting of this member is performed using the bed ways as a template. However, before any scraping or alignment is attempted, the spindle of the head stock must be given a series of accuracy tests.

Sec. 26.58

Accuracy Tests on the Headstock Spindle

These tests are designed to learn the condition of certain important features of the headstock spindle. Strictly speaking, the check-ups are not the duty or concern of the scraping operator, but for his own information and protection, he should conduct them. It is very possible that, if the spindle cannot pass one or more of these accuracy tests, it is so inaccurate that consistent readings could not be obtained, if the actual alignment procedures, accompanying the scraping operations, were to be executed.

Four accuracy tests are made on the spindle and will be described briefly. Obviously, they are not all of the examinations which the spindle would undergo during a manufacturing process, but they do constitute the indispensable ones which the scraping operator can perform conveniently and in a short time. The checks may be conducted with the headstock placed either on the lathe bed, or perhaps on a work bench.

Sec. 26.59
Spindle Center Runout (1st test)

PROCEDURE:

The test is begun by inserting a perfectly finished, live center into the tapered hole of the headstock spindle. The contact point of a DIAL INDICATOR is placed midway on the top of the 60° point of the live center. The spindle center runout is checked by revolving the spindle slowly.

This test will supply information on a number of matters, among them the correctness of the taper in the spindle; possible presence of burrs in the tapered hole; and concentricity of the axis of the tapered hole to the axis of the spindle. The tolerance shown in Fig. 26.55 cannot be exceeded.

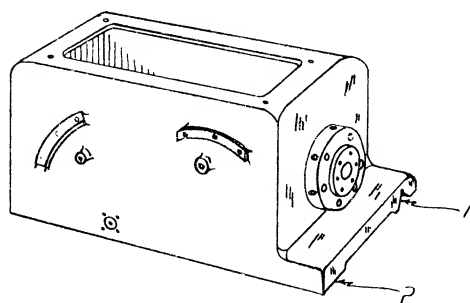


Fig. 26.54 View of the head stock of lathe
(1) V-slide (2) flat slide

Sec. 26.60
Spindle Nose Runout (2nd test)

PROCEDURE:

In this test the DIAL INDICATOR is positioned on the nose of the headstock spindle as shown in Fig. 26.56. The actual position it should occupy, whether directly on the nose, or indirectly, by attaching an accurate test plate, will depend upon the type of chuck mounting used, i.e. threaded, taper, or cam lock.

This test will check the concentricity of the nose with the axis of the spindle. Any error noted must be within the limits of the tolerance.

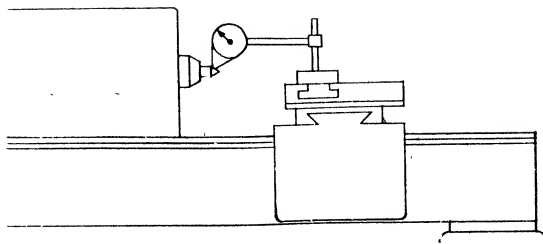


Fig. 26.55 Spindle center run out.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
(Total indicator reading)		
0 to .0004 "	0 to .0005 "	0 to .00075 "

Sec. 26.61
Cam Action of Spindle (3rd test)

PROCEDURE:

The cam action of the spindle is determined in the following manner. An accurately machined test plate similar in design to the back plate of a chuck is attached. The DIAL INDICATOR button is positioned on the rear side of the test plate, as shown in Fig. 26.57. The tolerance is also indicated. For supplementary material on Cam Action refer to Sec. 16.11.

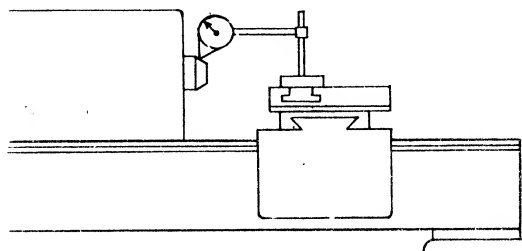


Fig. 26.56 Spindle nose run out

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
(Total indicator reading)		
0 to .0003 "	0 to .0004 "	0 to .0006 "

Having ascertained the accuracy of the spindle, the next step is to check the headstock spindle tapered hole.



PLATE 32. Inspector testing alignment of headstock spindle with test bar inserted in tapered hole. Readings are taken from DIAL INDICATOR mounted in lathe tool post. (Courtesy - South Bend Lathe Works.)

Sec. 26.62 Spindle Taper Runout (4th test)

There is a double purpose in conducting this test, namely:

1. To assure that the test bar is properly seated in the tapered hole of the spindle. It is necessary to know this so that readings obtained subsequently in aligning the headstock may be depended upon. In other words, the operator must be certain that the test bar when seated contributes no error to the readings which is not accounted for. (See Sec. 15.7 Sag in Test Bars)
2. To check the concentricity of the taper hole with the spindle axis. Incidentally, the concentricity is more accurately gaged by this set-up than by the test for Spindle Center Runout, due to the greater extension of the test bar.

We will now prepare the necessary apparatus for checking these two conditions.

A test bar is made up with a taper on one end to fit the tapered hole in the spindle. In making the bar, the straight portion should be proportioned to the size of the lathe and therefore will vary in length and diameter.

Before the bar is to be used its accuracy should be verified by mounting between centers and testing for concentricity. (See Sec. 15.11)

Next the bar is inserted into the tapered hole of the spindle. After mounting a DIAL INDICATOR on the carriage, the button is brought to bear against the test bar at say the vertical diameter. As the spindle is slowly revolved, readings are taken at both ends of the bar in turn. The readings obtained must be within the limits permitted, as indicated in Fig. 26.58. If this is accomplished, the test bar is known to be accurately seated in the spindle tapered hole.

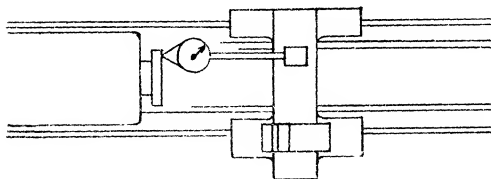


Fig. 26.57 Cam action of spindle

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
(Total indicator reading with instrument on rear side of test plate)		
0 to .0003"	0 to .0005"	0 to .00075"

Difficulty in accomplishing a zero-zero reading may be traced to grit or dust particles adhering to the surfaces. The safe practice is to clean the tapered hole with a cloth and stick. Never insert a finger in a revolving spindle.

As to the second consideration, the tapered hole must also be concentric with the axis of the spindle, within the tolerance indicated in Fig. 26.58. If the tolerance is exceeded there are several possible causes therefor.

To fix responsibility, the tapered section of the test bar should be examined to determine if the taper matches the spindle hole. This operation is easily performed by spreading a thin film of marking compound in a narrow streak, lengthwise on the tapered portion of the bar. By inserting the bar in the spindle hole and wringing it, some of the compound is transferred. Examination of the bar after

removing it will reveal if the streak is now spread uniformly around the entire tapered portion of the bar. This procedure will also disclose the presence of burrs, if any, in the spindle hole.

As the test bar is known to be accurate following the test between centers, and since faulty seating due to grit has been eliminated as a cause by careful cleaning, then possibly a non-uniform marking signifies that the tapered hole in the spindle may be affected by out of roundness, burrs, worn areas etc. Correcting this would, of course, entail a machining operation.

Having eliminated both the tapered hole and the test bar as possible factors of error, the operator may proceed with the headstock alignment tests. There is now assurance that the readings obtained henceforth will represent only the actual amount of misalignment of the headstock to the bed ways.

RECAPITULATION:

The purpose of the four tests just concluded, is to demonstrate the accuracy of the headstock spindle. The findings must show that any inaccuracy present in the spindle does not exceed the tolerances permitted in the several tests. If the tolerances are exceeded in one or more cases, the spindle should be reworked by grinding or machining, until it is capable of passing all four tests. The procedure to be followed in reworking the spindle is outside the scope of this book and will, therefore, not be discussed.

Sec. 26.63

Alignment Tests Conducted on the HEADSTOCK SPINDLE.

Three alignment tests will now be conducted on the headstock spindle. They have for their purpose the alignment of the headstock.

OBJECTIVES: Alignment of HEADSTOCK to the Bed.

1. Axis of the headstock spindle to be parallel to the outside ways of the bed in the vertical and horizontal planes.
2. Axis of the headstock spindle to be aligned with the axis of the tailstock spindle, in the vertical plane.

3. Slides to be fitted to the bed ways with a surface quality of 10 - 15 bearing spots per square inch.

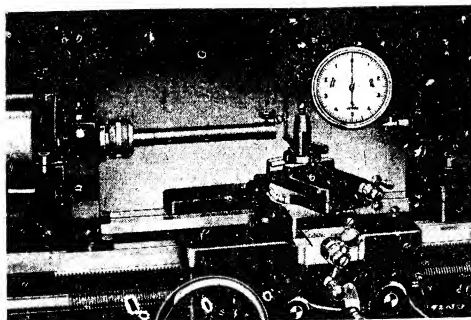


PLATE 33. Close-up showing application of DIAL INDICATOR for testing alignment of headstock spindle in vertical plane. (Courtesy - South Bend Lathe Works.)

PROCEDURE:

We will now by a series of tests determine approximately the degree of misalignment between the axis of headstock spindle and 1) DATUM PLANE and 2) axis of tailstock spindle. Apropos to this the headstock can, in the early stages of testing, be set on the bed without clamping. The final tests, though, should be conducted with it bolted solidly in position to obtain maximum precision.

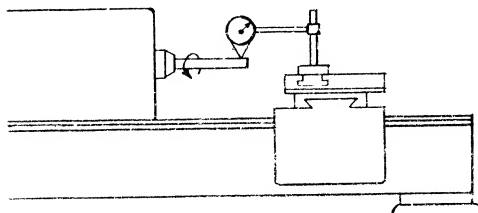


Fig. 26.58 Spindle taper run out.

RECOMMENDED STANDARDS

Tool Room Lathe	12" to 18" inc. Engine Lathe	20" to 36" inc. Engine Lathe
(Total indicator reading at end of 12" test Bar)		
0 to .0006"	0 to .0008"	0 to .00125"
(At end of spindle nose)		
0 to .0003"	0 to .0004"	0 to .0006"

Before performing the preliminary check for OBJECTIVE NO. 1 in the vertical plane, as shown in Fig. 26.59, a

spindle taper runout test is conducted. This requires the test bar to be inserted into the headstock spindle. Incidentally, a runout test must be performed each and every time the test bar is removed and re-inserted to prove the truth of the seating. To nullify the effect of eccentricity error, the mean position is located at the vertical diameter for this portion of the test.

Now returning to the alignment we will mount a DIAL INDICATOR on the carriage. The button of the instrument is placed at the vertical diameter of the test bar. Next the carriage is moved and a reading is taken between the spindle nose and the free end of the test bar. The amount of misalignment of the spindle to the bed ways in the vertical plane is thereby noted. This information relates to the first portion of OBJECTIVE NO. 1.

A similar preliminary check for OBJECTIVE NO. 1 in the horizontal plane is also necessary. This requires finding the mean position of the test bar and turning it to the horizontal plane. Then the button of the instrument is set at the near side of the test bar i.e. the horizontal diameter, as shown in Fig. 26.60. Again the carriage is traversed and the reading is recorded.

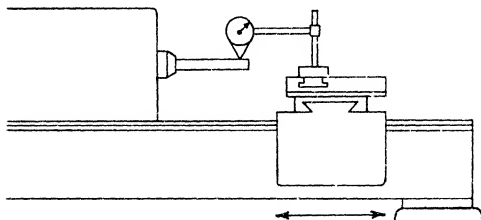


Fig. 26.59 Alignment of head stock spindle in vertical plane.

RECOMMENDED STANDARDS

Tool Room Lathe	12" to 18" inc. Engine Lathe	20" to 36" inc. Engine Lathe
0 to .0005"	0 to .001"	0 to .001"
(high at end of 12" test bar)		

These two tests will provide necessary information as to the degree of misalignment now existent in the headstock spindle with respect to the DATUM PLANE i.e. the outside inverted V-ways of the bed. This data refers only to OBJECTIVE NO. 1.

Inasmuch as all three OBJECTIVES must be attained simultaneously, it is important to know at this stage, before any scraping is attempted on the headstock

slides, the approximate misalignment in the vertical plane of the headstock spindle with respect to the axis of the tailstock spindle. Since the headstock surfaces have not as yet been treated, it is obvious that the axis of the headstock spindle is higher from the bed ways than the axis of the tailstock spindle, which member has been scrape-finished. We determine approximately how much metal to remove from the headstock V-slide and flat slide by executing the following procedure:

Sec. 26.64

Vertical Alignment of HEAD & TAIL CENTERS

The necessary equipment for this operation includes a test bar of a different type than those previously used. A hardened steel cylinder of suitable length, centered at both ends and then accurately ground to one appropriate overall diameter will be required. It will be mounted between the lathe centers during the test.

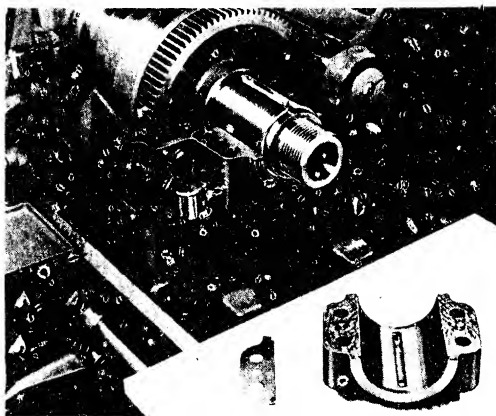


PLATE 34. Close-up of headstock with spindle bearing cap removed to show how surface of headstock and bearing cap are hand scraped and fitted together. (Courtesy - South Bend Lathe Works.)

To facilitate the test for vertical alignment it is necessary to adjust, as closely as possible, the axis of the tailstock spindle in-line with the axis of the headstock spindle in the horizontal plane. The preliminary adjustment is accomplished by the following set up:

After fastening a DIAL INDICATOR to the carriage the contact button is

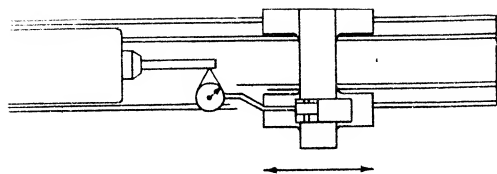


Fig. 26.60 Alignment of head stock spindle in horizontal plane.

RECOMMENDED STANDARDS

Tool Room Lathe	12" to 18" inc. Engine Lathe *	20" to 36" inc. Engine Lathe
0 to (+) .0003"	0 to (+) .0005"	0 to (+) .0008"

(At end of 12 inch test bar)

placed on the horizontal diameter of the test bar, as shown in Fig. 26.61. The carriage is now moved so that the button slides from end to end of the bar. At the same time the tailstock top is set-over on its transverse way until a reading close to zero-zero is attained. We say "close" because at this stage we would not obtain an identical calibration in traversing the test bar from end to end due to the fact that the axis of the headstock spindle is higher than the axis of the tailstock spindle. The reason, as previously stated, is that no scraping has yet been done on the slides of the headstock. Consequently, the DIAL INDICATOR button would ride up or down from the horizontal axis of the test bar as the carriage is moved along the bed ways.

Having adjusted the bar closely, we can proceed to check the vertical alignment. To this end the DIAL INDICATOR is positioned on the vertical diameter of the test bar, as shown in Fig. 26.62. Readings are now taken as the carriage is moved along the bed. The results of this test indicate approximately how much metal must be scraped from the bearing surfaces of the headstock to obtain correct alignment and satisfy OBJECTIVE NO. 2.

After the approximate amount of required correction is learned from this series of tests, spotting and scraping

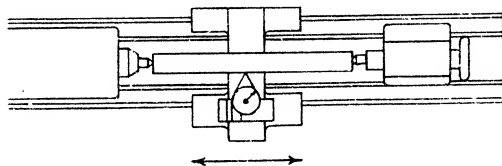


Fig. 26.61 Set up utilized to adjust headstock and tailstock centers in horizontal plane.

operations can be initiated. The headstock is placed carefully on that portion of the bed ways previously coated with marking compound. Then after spotting, the member is removed from the bed and laid on a work bench where it is scraped. Since it is unlikely that a single cycle of spotting and scraping will accomplish the alignments specified in OBJECTIVES NO. 1 and NO. 2, it will be necessary to repeat the foregoing series of tests a number of times. As the alignment approaches the required accuracy, the adjustment of the tailstock in the horizontal plane will be progressively smaller. The need for it will end when the DIAL INDICATOR reading nears zero-zero on both the horizontal and vertical diameters.

When the tests disclose that the correct alignment is imminent, the desired bearing quality is worked into the headstock V-slide and flat slide, to satisfy OBJECTIVE NO. 3. Refer to Sec. 23.25. Simultaneous Achievement of Bearing Quality.

If the tolerances indicated in the tests above mentioned, are not exceeded in the final alignment of the headstock spindle, then the three OBJECTIVES are satisfied and the headstock member is considered completed. However, if during the course of trying to obtain the OBJECTIVES, the scraper should remove too much metal from the slides, thus lowering the axis of the headstock spindle below the axis of the tailstock spindle, it is obvious that OBJECTIVE NO. 2 cannot be satisfied.

Faced with this dilemma, the scraper should continue scraping until OBJECTIVES NO. 1 and NO. 3 are achieved at which moment the headstock is adjudged completed. OBJECTIVE NO. 2 is disregarded.

The matter of the vertical alignment cannot, however, be dismissed. Rectification, in such cases, is obtained by returning to the tailstock. The slides of this member must be reworked, to satisfy all the OBJECTIVES listed in Sec. 26.55, plus this additional one, viz:

"Axis of tailstock spindle to be aligned to axis of headstock spindle in the vertical plane."

When these objectives are satisfied then both of the members are correctly aligned, to each other and to the bed ways. Thus the accuracy of the lathe is in nowise impaired by failure or inability to achieve alignment by means of the headstock.

The correct alignment can still be obtained by reworking the slides of the tailstock.

There is an alternative method of checking the alignment of the axis of headstock spindle to axis of tailstock spindle in vertical plane, as specified in OBJECTIVE NO. 2. The principal advantage of this system is that alignment of the axes of the two centers in the horizontal plane is not required. Another advantage in substituting it is that checking the alignment after each scraping cycle can be done more speedily. A further advantage of the method is that the exact degree of misalignment can be ascertained, not just the approximate amount.

The necessary apparatus for applying the method includes a DIAL INDICATOR mounted on the cross-slide on the carriage. Two precision ground test bars of identical diameter and length are also used. One is inserted in the spindle of headstock, the other is inserted in spindle of tailstock. Both bars are given their usual tests for proper seating, runout etc. Then the button of the DIAL is adjusted to the vertical diameter of each bar at the free end in turn. This can be done without changing the setting of the instrument by moving the cross-slide out and in. Readings are taken which, if within the tolerance allowed, signify that the axis of both spindles are truly aligned in the vertical plane.

NOTE:

During the course of aligning other components of the lathe, the movable members will many times be slid back and forth on the ways of the bed. This causes isolated high spots which previously escaped notice to become sharply defined. Therefore, when all alignments have been completed on the other members, it is good practice to give the bed ways a final touching up to improve the wearing quality, that is, to make it more lasting. (Refer to Sec. 19.12)

Bronze Bearings in Spindles

As a general rule, the slides of the lathe headstock do not wear or warp since the member is bolted solidly to the bed. Consequently, treatment of these surfaces is necessary only for alignment purposes. That is, as the discussion in Sec. 26.63

and Sec. 26.64 has shown, the axis of the headstock spindle must be parallel horizontally and vertically to the bed ways and in-line with the axis of the tailstock.

If the headstock spindle has bronze bearings it is practicable to secure these essential alignments by scraping the round bronze bearing instead of the flat slides. This operation is performed with the headstock bolted to the bed ways. Scraping is done with a round-scraper. The several alignment tests are executed periodically.

Sec. 26.65

Final Alignment of the Carriage Ways

The OBJECTIVES which were sought in scraping the compound slide rest and the carriage were designed so that the member would be correctly aligned to the headstock spindle. How exactly the work was done will be determined by the following alignment test.

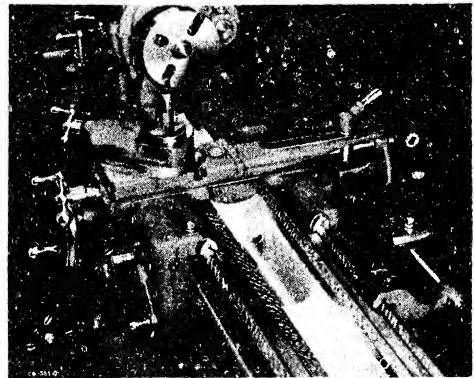


PLATE 35. Close-up of lathe carriage and headstock showing how the units are aligned on the lathe bed by inverted V-ways which are hand scraped straight and parallel. Note taper attachment at right side of illustration. (Courtesy - South Bend Lathe Works.)

OBJECTIVES: The FINAL ALIGNMENT of the CARRIAGE WAYS to the AXIS of HEADSTOCK SPINDLE.

1. Cross slide movement to be square with axis of headstock spindle.
2. Flat ways to be level, longitudinally and transversely.
3. Carriage V-slides to be fitted to bed ways with a surface quality of 10 - 15 bearing spots per square inch.

PROCEDURE:

To determine the variation from correct alignment, the following test is now conducted. Incidentally, any set up which will accomplish the same purpose as the one here described may be used. In the present case a four jaw chuck is attached to the headstock spindle and a steel PARALLEL is then lightly clamped in the jaws, as illustrated in Fig. 26.63. The PARALLEL is adjusted so that approximately half of its length extends on either side of the spindle axis. Position the cross slide on the carriage, insert the cross slide gib and adjust for sliding pressure. Attach the gib bracket and adjust the carriage gib for sliding pressure. Next clamp the carriage to the bed.

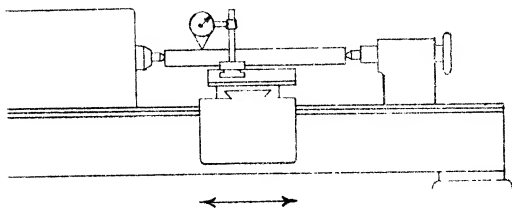


Fig. 26.62 Alignment of head and tail centers in vertical plane.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
0 to .0008"	0 to .001"	0 to .0015"
(High at tailstock end)		

A DIAL INDICATOR is now mounted on the cross slide and placed in contact with the PARALLEL near one end, for example, at (A) in Fig. 26.63. Revolve the spindle, meanwhile adjusting the PARALLEL by tapping it. A zero-zero reading must be registered by the DIAL as each end (A) and (B) of the PARALLEL comes in contact with the plunger button of the instrument. When this is accomplished, the PARALLEL is square with the axis of the spindle.

Now turn the spindle until the steel PARALLEL is approximately level. Maintained in this position, the button of the device will not ride off as the slide rest is moved in and out transversely.

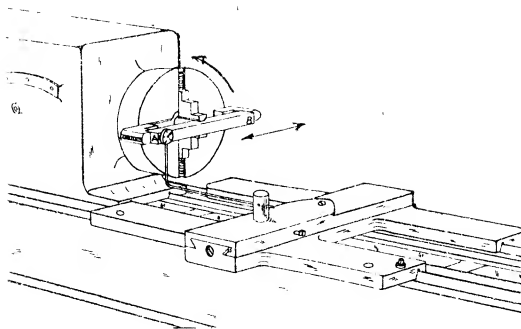


Fig. 26.63 Showing method of adjusting PARALLEL square to axis of spindle. Revolve spindle and adjust PARALLEL until a zero-zero reading is obtained as each end of the PARALLEL makes contact with DIAL INDICATOR.

Since a unilateral tolerance is allowed, as shown in Fig. 26.64, the DIAL can have a minus reading only, when moving the cross slide from the front side of the lathe towards the center. Conversely, it can have a plus reading only, when moving away from the center, towards the front side of lathe. If the tolerance is not exceeded, the transverse movement of the cross slide is satisfactory. However, if it does not pass this test, then the two V-slides of the carriage must be scraped so as to "throw" the cross slide in the correct direction and by the proper amount. When this test is passed, OBJECTIVE NO. 1 is satisfied.

By occasionally laying a PRECISION LEVEL on the flat ways in both the longitudinal and transverse directions, as shown previously, notably Fig. 26.30 and Fig. 26.32, a close check can be kept on OBJECTIVE NO. 2. This is essential because there is a tendency to tilt these surfaces if the scraping performed on the carriage slides is carelessly executed.

IMPORTANT:

It might be worthwhile to emphasize that, although we are aligning the carriage ways, the correction is made on the carriage slides.

The surface quality required by OBJECTIVE NO. 3 should be produced simultaneously.

NOTE:

Any rescraping of the carriage V-slides may also necessitate reworking the rear carriage gib bracket bearing

to a correct alignment with the rear carriage gib way of the bed. Obviously, the gib piece will also be affected by this change and may require extensive rescraping depending upon the amount of error disclosed.

Sec. 26.66

OBJECTIVES: Final Alignment Test of Compound Rest Top.

- 1. Movement of compound rest top to be parallel to headstock spindle in vertical plane.

PROCEDURE:

The compound rest is used for machining or boring a steep taper and unless the tool is exactly on-center throughout its entire travel, a true taper will not be generated. In short, the surface lines of the work piece will be hyperbolic instead of straight and tapered. It is to insure this condition that we make a final test on the compound rest top, after the carriage V-slides have been scraped so as to correctly align the movement of the cross slide.

Returning to the procedure, the test bar is inserted into the headstock spindle and, after the customary runout is performed, revolved to place the mean position of eccentricity error in the vertical plane. A DIAL INDICATOR mounted on the compound rest top should be adjusted to permit the INDICATOR button to touch the test bar at the vertical diameter.

The compound rest is swiveled so that the movement of the compound rest top parallels the bed ways. By turning the feed screw, the compound rest traverses, sliding the button along the test bar.

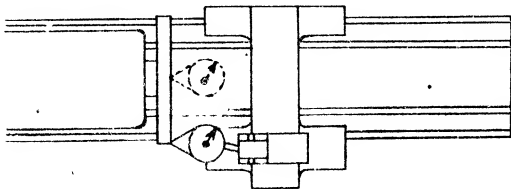


Fig. 26.64 Testing alignment of cross slide movement to axis of headstock.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc.	20" to 36" inc.
	Engine Lathes	Engine Lathes
0 to .0005"	0 to .001"	0 to .001"
(To face hollow or concave only on 12" diameter)		

The tolerance allowed is shown in Fig. 26.65. If this is exceeded, one possibility could be that the flat ways of the compound rest swivel were not scraped parallel with the swivel slide. Sec. 26.20 discusses a method of dealing with this.

This completes all scraping operations on the lathe. The feed screws of the cross slide and compound rest may now be checked.

Sec. 26.67

Back Lash on Feed Screws

PROCEDURE:

The accompanying chart Fig. 26.66, discloses the recommended standards for the back lash on the cross feed screw and on the compound rest feed screw.

The feed screws are tested for back lash, while threaded into their respective nuts.

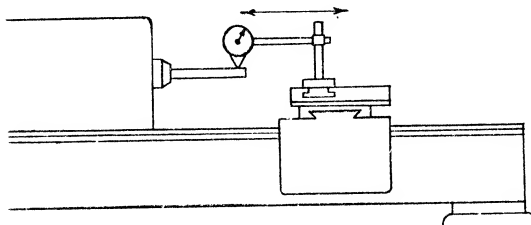


Fig. 26.65 Final alignment of compound rest top. Movement of slide parallel with work spindle in vertical plane. Tol. max. .0035" per foot.

If the tolerance is exceeded, the nut should be replaced or, if of the compensating kind, adjusted to take up for wear.

Now that the alignments of the lathe members are finished, the next operation is the assembly of miscellaneous unattached parts. Strictly speaking, this is not scraping work, but it may nonetheless be the responsibility of the operator.

Sec. 26.68

Assembly of Lead Screw and Apron

In assembling the lead screw and apron, one point to watch is the adjustment that may be required between the lead screw bearings and the half nut to accomodate them to each other under the new and changed conditions. Since more or less metal has been scraped from both the bed ways and the carriage slides, it is obvious that the present position of the half nuts

will be lower in relation to the lead screw than formerly. Likewise the rack may have to be lowered to prevent excessive play in the handwheel. Adjustments of this nature, though essential, do not pertain to the subject matter of this book and therefore will not be discussed further. However, the importance of the LeadScrew in the efficient operation of the machine justifies the space allotted to several tests concerning its alignment to the halfnut and the bed ways. These matters will now be discussed.

Sec. 26.69
Lead Screw Alignment.

PROCEDURE:

Due to the removal of metal from various bearing surfaces, the lead screw will be misaligned if replaced without adjustment of its bearings. For proper adjustment the lead screw must be lowered to obtain a correct alignment with the half nuts. In addition, the lead screw must be aligned to the bed ways.

To test this alignment the lead screw is inserted in the bearings. The carriage is placed at the mid-point on the bed ways, closing the half nuts on the lead screw. A simple indicating jig, or the one shown in Fig. 26.67a, is then made up. When set in place it is guided by the front inverted V-way of the bed, and supported at the other end by the rear inverted V-way.

Next a DIAL INDICATOR is attached to the jig with the button adjusted to touch the lead screw at the vertical diameter. The apparatus is moved along the ways and checks are made at short intervals from end to end of the lead screw. This is repeated with the button at the horizontal diameter. The tolerance allowed is shown in Fig. 26.67b. If exceeded, the lead screw bearings are replaced or shimmed to obtain the correct alignment.

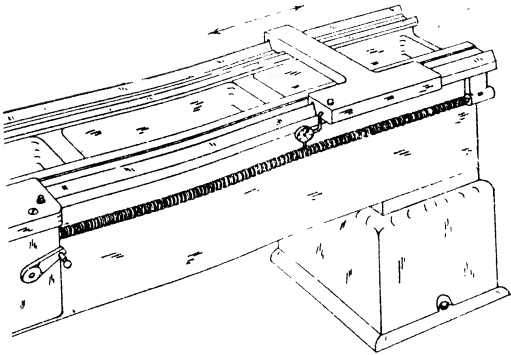


Fig. 26.67(a) General arrangement for testing lead screw alignment with bed ways.

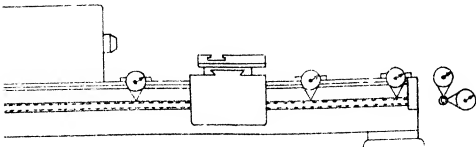


Fig. 26.67(b) Alignment of lead screw

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
0 to .004 "	0 to .004 "	0 to .006 "
(Parallel with ways horizontal and vertical)		
0 to .006 "	0 to .006 "	0 to .008 "
(Alignment of half nut horizontal or vertical)		

Sec. 26.70
Lead Screw Cam Action

PROCEDURE:

This test is designed to show the maximum permissible cam action of the lead screw. Fig. 26.68 shows the set up which should be self explanatory. The tolerance is also indicated.

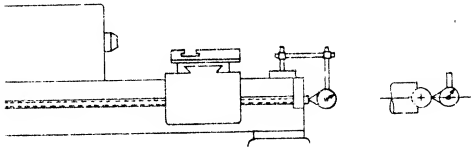


Fig. 26.68 Lead Screw Cam Action

Fig. 26.66 (a) Back Lash on Cross Feed Screw
(b) Back Lash on Compound Rest Screw

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
(a) .004 "	.004 "	.005 "
(b) .004 "	.004 "	.005 "

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathe	20" to 36" inc. Engine Lathe
max. .0003 "	max. .0004 "	max. .0005 "

Sec. 26.71

The Accuracy of Lead Screws

The accuracy of lead screws should meet the recommended standards shown in the accompanying chart Fig. 26.69. In

Fig. 26.69 Accuracy of lead screw

RECOMMENDED STANDARDS		
Tool Room Lathe	12" to 18" inc. Engine Lathe	20" to 36" inc. Engine Lathe
($\pm$) .001" (lead per foot)	($\pm$) .0015"	($\pm$) .002"
($\pm$) .0004" (lead in any 4")	($\pm$) .0005"	($\pm$) .0007"

reality, this is a manufacturer's test and most shops would not be concerned with this phase. Several methods of testing the pitch error in lead screws will be provided, though, for the benefit of those interested.

One application utilizes center point combinations. In this operation PRECISION GAGE BLOCKS, mounted in a holder, provide an accurate method for checking the pitch of the lead screw threads.

Another method employs optical instruments for comparative testing of the lead screw against a precision test bar.

A third procedure uses the travel of the carriage to test the pitch error. This method will be discussed at some length because it can be performed with equipment readily available in the average shop.

Sec. 26.72

Pitch Error in Lead Screws

(The accuracy tests of lead screws is based on a reference length of 12" with a further subdivision of 1/3 or 4" .)

PROCEDURE:

This procedure involves taking the measurements over a distance of 12". Starting at any distance from the headstock, the half nut is closed on the lead screw. Then the carriage is moved along the bed by turning the lathe spindle by hand. A count is made of the total number of revolutions which theoretically should advance the carriage the standard 12".

Two measurements are executed. The first, is made in conjunction with a fixed type end gage, as shown in Fig. 26.70a. This determines the starting position.

The second measurement is performed without the fixed type end gage, as shown in Fig. 26.70b. This is the stop position.

Since the figures are nearly self-explanatory we shall call attention here to only a few salient features. The set up is arranged as follows:

A ground stud (1) is bolted to the face plate. A sleeve (2) having a close sliding fit is slipped over the stud. Next a bar (3) is clamped to the top of the lathe bed. To the bar is fastened a DIAL INDICATOR (4). The DIAL button should be placed in contact with the sleeve. Some discretion is required when choosing the contact position, so that when the sleeve is removed, the stud will clear the DIAL INDICATOR as the lathe spindle is turned.

The lathe carriage is stationed on the bed at the point where the lead screw is to be tested. Another DIAL INDICATOR (5) is attached to the carriage. A stop (6) is clamped approximately 12" from the INDICATOR (5). A fixed type end gage (7) is made up to exactly 12" in length. It is positioned in a PRECISION V-BLOCK (8) laid on the inverted V-way.

After closing the half nuts all slack is taken up in the driving mechanism by turning the spindle until the button of the DIAL INDICATOR (5) is in contact with the fixed end gage (7). This INDICATOR reading is jotted down for later reference.

The DIAL INDICATOR (4) is now positioned against the sleeve (2) with sufficient pressure to register on the DIAL, and this reading, too, is noted. The sleeve (2), the end gage (7), and the PRECISION V-BLOCK (8) are removed. Following this, the spindle is turned the exact number of times which mathematical calculations show would traverse the carriage 12". The actual variation over or under represents the Pitch Error.

To illustrate this procedure let us assume an imaginary example:

Assuming the lead screw has eight threads per inch, the pitch of thread would then be 1/8". Therefore, the spindle must be revolved eight times to move the carriage one inch, or ninety six times to traverse the carriage the required 12". The actual travel of the carriage will

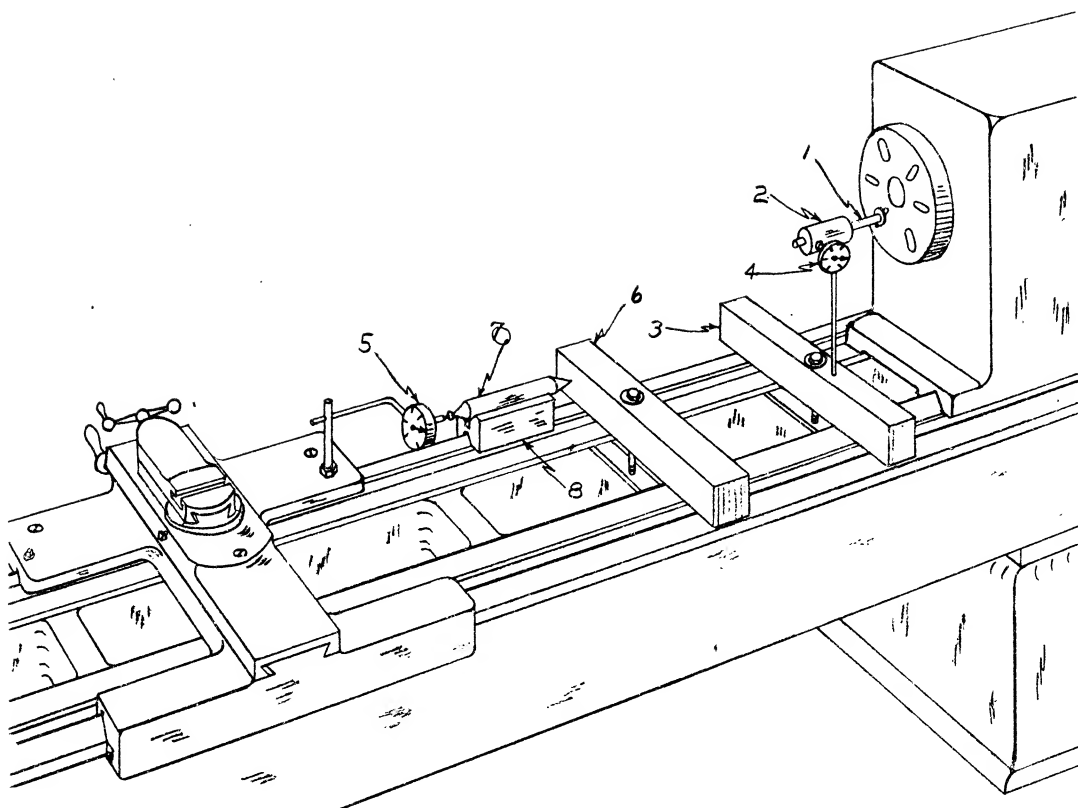


Fig. 26.70(a) Testing pitch error in Lead Screw by comparison with End Gage. 1st phase Starting Position.

(1) ground stud (2) sleeve (3) parallel bar (4) DIAL INDICATOR
(5) DIAL INDICATOR (6) parallel bar (7) end gage (8) precision V-block.

be either more or less than 12" and this difference when measured will be the pitch error.

In practice, the spindle is revolved ninety five times and as the last turn begins, the sleeve is inserted on the stud. The ninety sixth revolution is completed when the reading on the INDICATOR (4) attains the same value as previously recorded. At that point further movement should be stopped.

We now turn to the INDICATOR (5). The button of this DIAL should be in contact with the stop (6). Whatever the difference between the first reading which was jotted down and the present reading will constitute the pitch error of the lead screw in thousandths of an inch per foot, either plus or minus.

Sec. 26.73

Limitations Affecting the Accuracy of Static Alignment Tests.

With the lathe completely assembled it is well to try the lathe under actual operating conditions. Up to this point we have depended principally upon a DIAL INDICATOR to inform us as to the accuracy of our alignments. Furthermore, no working pressure has been applied to the bearing surfaces. However, we may feel reasonably certain that the machine will produce accurate work, provided the following conditions were fulfilled, namely:

1. If an accurate, sensitive DIAL INDICATOR was used.
2. If testing set-ups were made accurately.

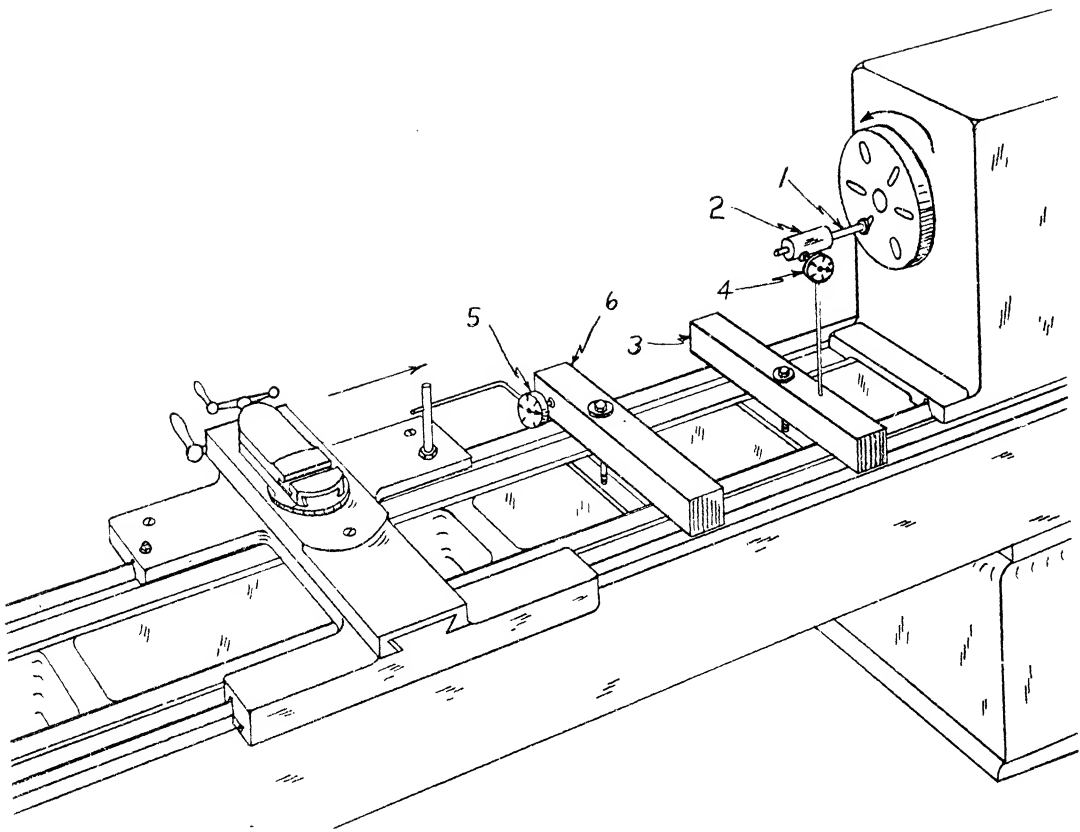


Fig. 26.70(b) Testing pitch error in lead screw. 2nd Phase. Stop Position. The carriage has been advanced 12 inches more or less in direction of Arrow. Any variation in the reading of DIAL attached to the carriage indicates the pitch error.

3. If the tests were conducted so as to obtain pertinent information. Unless clear thinking is the order, it is entirely possible to perform spurious tests which prove nothing and have no significance or application to the problem.
4. If a careful and workmanlike job of scraping was done.
5. If room temperatures did not vary widely during the period when alignment tests were made.
6. If all surfaces of members in contact were free of dirt, grit, and thick coatings of marking compound or films of heavy oil.
7. If indicating jigs were rigidly made and properly applied.
8. If SURFACE PLATES, Templates, and other spotting tools were in perfect condition.

When any of these factors has been

slighted by the scraping operator, or if machine repairs on spindles, gearing etc. were performed in a slip shod fashion, then we are almost certain to experience trouble in securing satisfactory results under actual operating conditions. To dismantle the lathe again, to correct some individual fault, is definitely not good practice. Furthermore, it suggests the possibility that other dependent surfaces or alignments may also require rectification.

It is always good practice to run in the machine under power for about one hour. During this warming-up period, the various members are repeatedly tested for quietness and smoothness of movement. When the machine attains a constant, normal, operating temperature the Final Tests under Working Conditions may be performed.

Tests Under Operating Conditions

Now we will proceed to tests under actual working conditions.

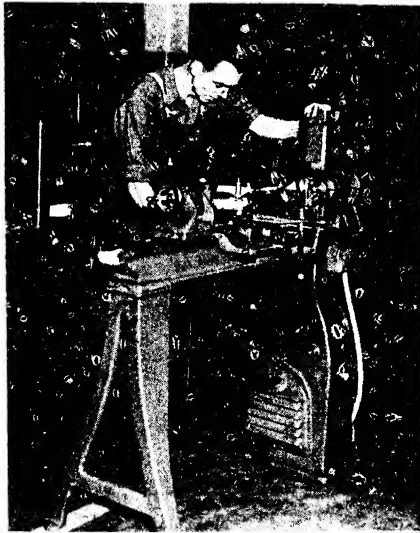


PLATE 36. Inspector taking a trial cut on a test piece held in lathe chuck to test truth of headstock. (Courtesy - South Bend Lathe Works.)

Sec. 26.74

Lathe Must Turn Round with Work Mounted in Chuck.

PROCEDURE:

A short bar of steel, two or more inches in diameter, is placed in the chuck. A skim cut is taken on the bar, following which it must check round within the tolerances given in Fig. 26.71. If excessive

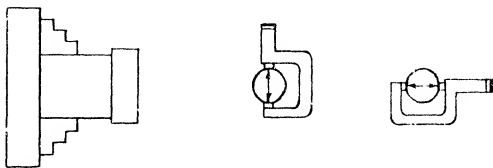


Fig. 26.71 Lathe must turn round with work mounted in chuck.

RECOMMENDED STANDARDS

Tool Room	12" to 18" inc.	20" to 36" inc.
Lathe	Engine Lathe	Engine Lathe
.0003"	.0004"	.0008"

out-of-roundness is noted, an examination of the headstock main bearings and spindle should reveal the cause of the inaccuracy.

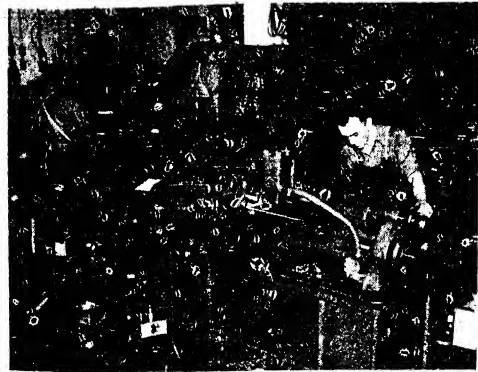


PLATE 37. General view in factory showing inspector at work testing the alignment of the headstock spindle on the lathe bed. Note the test bar in spindle of lathe to left rear. (Courtesy - South Bend Lathe Works.)

Sec. 26.75

Lathe must turn Cylindrical with Work Mounted in Chuck.

PROCEDURE:

A length of steel bar stock of suitable diameter is mounted in the chuck. Two collars are machined on the work piece; one collar near the chuck jaws, the other collar at the free end of the bar.

The next step is to take a very light finishing cut across one collar, and without changing the adjustment of the tool, the cut is continued across the other collar. The diameter of each collar should be the same, otherwise a taper is indicated. The tolerance allowed is shown in Fig. 26.72.

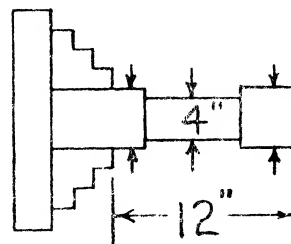


Fig. 26.72 Lathe must turn cylindrical with work mounted in chuck.

RECOMMENDED STANDARDS

Tool Room	12" to 18" inc.	20" to 36" inc.
Lathe	Engine Lathe	Engine Lathe
.0008"	.0015"	.002"

There are several possible causes if a taper is in evidence, namely:

1. The lathe is no longer level. This is an unlikely possibility if the lathe has not been moved from the location where it was reconditioned. It is, however, a simple matter to check the levelness of the lathe to make certain that the shims have not shifted.
2. The ways have a "wind". If the outer inverted V-ways of the bed were scraped in a slipshod manner it is possible they are not straight. They must be checked to determine their accuracy. Pertinent tests would include, the Microscope & Taut Wire System, Water Level System.
3. The headstock is misaligned. This means that the headstock spindle may not be parallel to the DATUM PLANE. Repeating tests given in Sec. 26.63 and shown in Fig. 26.59 and Fig. 26.60 will determine if the error can be charged to this component. In order to correct any misalignment discovered, the entire scraping and testing procedure commencing with Sec. 26.63 must be repeated because related members were dependent upon the alignment of the headstock spindle and they will now be out of true.

(As an alternative to machining two collars, the accuracy of alignment of the headstock spindle may be determined by boring a hole with the work piece held in the chuck. If a taper develops, the source of error may be located in one of the possibilities outlined above.)

Sec. 26.76

Lathe must turn cylindrical with the Work mounted between centers.

PROCEDURE:

A length of cylindrical bar stock of appropriate diameter is centered at both ends and placed between the lathe centers. Using a light cut and fine feed, two collars of the same or random diameter are machined on the bar. One of the collars is formed near the headstock center and the other adjacent to the tailstock center. Certain measurements will now be made on these collars to prove that the lathe can

turn out accurate work as specified above.

1. To determine roundness.

The collar at the headstock end, in particular, should be "miked" because there the error, if any, will be most pronounced. The collar is measured on the vertical and horizontal diameters and at intermediate points. If out-of-roundness is apparent the error may be attributed to the following sources:

- (a) Inaccurate spindle or spindle bearings. Refer to Sec. 26.74.
- (b) Spindle center run out. Refer to Sec. 26.59 and see Fig. 26.55.

These causes of out-of-roundness should be investigated and the fault corrected before searching for other possible errors.

2. To determine cylindricity:

Next a very light finishing cut is taken across both collars without changing the setting of the tool. Then the diameter of each collar is "miked" at several places. The tolerance allowed is shown in Fig. 26.73. The measurements of one collar

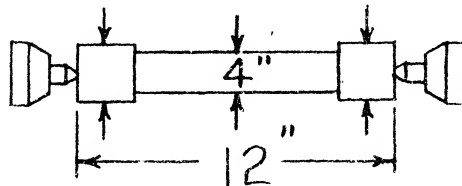


Fig. 26.73 Lathe must turn cylindrical with the work mounted between centers.

RECOMMENDED STANDARDS

Tool Room Lathe	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathe
.0004"	.0008"	.001"

must correspond to the measurements of the second, otherwise a taper is indicated. A taper may be due to the following reasons:

- (a) The bed is not level. To dispose of this possibility, repeat the leveling checks described in Sec. 26.38. The tolerance is given in Fig. 26.29a and Fig. 26.29b.

- (b) The axis of tailstock spindle is not in-line with axis of headstock spindle in horizontal and/or vertical planes. The following system is used to determine in which plane the error is located.

Any error caused solely by a misalignment in the horizontal plane can be eliminated by mere mechanical manipulation of the tailstock. First, "mike" the collars, and if the diameter of the collar at the tailstock end is the larger of the two, the lateral adjustment screws are turned to move the tailstock top in its transverse way towards the front of the lathe. On the other hand, if the collar at the tailstock end has a smaller diameter the tailstock is set over towards the back of the lathe. Repeated cuts, measurements, and adjustments are made until the diameters of both collars are identical. In this way all error in the horizontal plane is eliminated.

To determine if misalignment is present in the vertical plane, the following procedure may be adopted. This test is most easily performed by mounting a DIAL INDICATOR on the tool post and adjusting the button to touch one of the turned collars at the vertical diameter. The pointer is set to zero, and then without changing the setting, the carriage is traversed along the bed ways and the reading noted at the other collar. If there is no misalignment in the vertical plane, the reading will be identical. If there is a difference, correction is made by scraping the slides of either the headstock or tailstock, whichever has its axis higher as measured from the bed ways.

CAUTION:

The machining of the two narrow test collars of the same diameter, one at each end of the bar, at a single setting of the cutting tool, is not the infallible test for alignment of the axis of the tailstock spindle to axis of headstock spindle that it is popularly supposed to be. Should the tailstock spindle be either higher or lower than the headstock spindle, it is still possible to machine, without changing the setting of the tool, two narrow test collars of apparently identical diameters by simply adjusting the tailstock in its transverse way. The reason for this phenomenon

is that for any particular height of the cutting edge of the tool there will be a certain set-over of the tailstock which will apparently compensate for the misalignment of the tailstock axis in the vertical plane. In reality, the collars will be hyperbolic rather than cylindrical in character. Where the misalignment in the vertical plane is small, variations in micrometer readings on the several diameters will be almost imperceptible. Therefore, in order to make an accurate test of the alignment of the head and tail centers in the vertical plane, the following procedure should be adopted.

Machine two collars at opposite ends of a bar held between centers, using a light cut and fine feed. Measure the collars with a micrometer. If the measurements are not the same, adjust the tailstock transversely, out or in as required, and machine again. Machine and measure consecutively until the collars "mike" identically. Then fasten a DIAL INDICATOR to the tool post of the carriage and adjust the button of the instrument to the vertical diameter of one collar. Traverse the carriage along the bed and observe the DIAL reading at the other collar. If the reading is zero-zero the axes of the head and tail centers are correctly aligned in the vertical plane.

An additional test may be made which is based on the premise that if the headstock spindle and tailstock spindle are correctly aligned to each other in both planes, then the height of the cutting tool above, below, or on-center will produce identical diameters in the two test collars without changing the setting of the tool on the machining cut. Thus after obtaining equal diameters on the two collars by adjusting the tailstock, a second cut is made across both collars, this time with the cutting edge of the tool raised or lowered from its first setting. Any misalignment of the tailstock in the vertical plane will be immediately apparent by a difference in diameters of the two test collars.

Sec. 26.77
Lathe to Face Concave Only

PROCEDURE:

For this test the face plate is mounted on the headstock spindle. Lock the carriage and set the lathe cutting tool to take a light facing cut. After the cut is completed, the alignment of the guiding way of the carriage may be tested by means of the Three Papers Method. (Sec. 22.4)

To perform this operation, three paper slips are positioned in a row against the face plate. One paper is placed near the middle. The other two are oppositely located at the outer rim of the face plate. They are held there by pressing them with a hardened steel straight edge. By judging the drag of the center paper as it is removed, the operator determines if the tolerance shown in Fig. 26.64 has been exceeded. If this is the case the carriage ways are misaligned with respect to the axis of the spindle. Correction of the error calls for rescraping the carriage slides, as discussed in Sec. 26.65.

This concludes the tests made on the assembled lathe. If all tests were satisfactorily finished, the lathe, insofar as the scraping and alignment of major members are concerned, may be considered completed.

No mention has been made of minor components, such as taper attachments, special fixtures etc. But it is felt that if scraping or alignment is needed on these members, there is sufficient information available in these pages to assist the scraping operator when treating them.

Accepted standards of Accuracy for a few common attachments are included herewith to serve as a guide.

Sec. 26.78
Chuck Run Out

PROCEDURE:

The recommended standards of accuracy for chucks are illustrated in the accompanying Fig. 26.74.

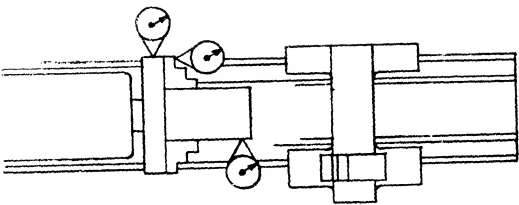


Fig. 26.74 Chuck run out.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathes
.003"	.003"	.004"
(face and periphery)		
.003"	.003"	.004"
(face of steps)		
.003"	.003"	.004"
(Bar test 3" from end of jaw. Bar diameter same as hole)		

Sec. 26.79
Collet Chuck Run Out

PROCEDURE:

The set up shown in Fig. 26.75 will indicate any variation from the recommended standards for collet chucks.

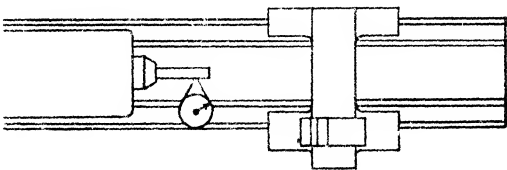


Fig. 26.75 Collet chuck run out.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathe
0 to .001"	0 to .001"	0 to .001"
(one inch from spindle)		

Sec. 26.80
Face Plate Run Out

PROCEDURE:

The accompanying Fig. 26.76 furnishes the accuracy standard for face plates.

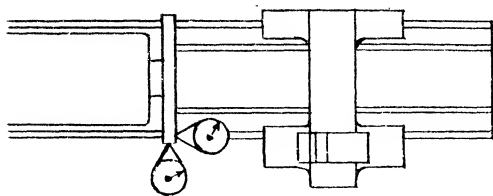


Fig. 26.76 Face Plate run out.

RECOMMENDED STANDARDS

Tool Room Lathes	12" to 18" inc. Engine Lathes	20" to 36" inc. Engine Lathe
0 to .0005"	0 to .001"	0 to .0015"
(on diameter)		
0 to .001"	0 to .0015"	0 to .002"
(on face at nominal diameter)		

Sec. 26.81
Effects of Inaccurate Leveling

Although the lathe was leveled prior to scraping, this procedure must be repeated again if the machine is moved to another place for permanent erection. Any relo-

cation of the bed necessitates the exercise of due care on the part of the user. If the machine is placed in operation when the lathe bed is not level, serious trouble will be experienced. The ways of the bed become twisted thereby initiating a whole series of unfortunate consequences. For example, when the ways of the bed are twisted due to faulty leveling the carriage slides cannot fit properly. Operated in this condition, chatter marks will be generated on the work piece.

A "wind" in the bed ways, attributable to improper leveling, will also twist the headstock out of line. This binds the spindle in its bearings. The bearings may score and deterioration is accelerated. Chatter marks are formed on the work piece by this cause also.

Improper leveling, besides impairing surface quality, likewise affects the accuracy of the work output in the following manner. Work held in a chuck cannot be turned straight, nor can a hole be bored which will not taper. It is also impossible to turn work straight when held between centers.

To function accurately, a lathe regardless of size, or weight must be perfectly level before being placed in use. It is the purpose of the leveling process to relieve strains and distortion and restore the bed to the plane in which it was scraped.